

Class11: Structural Bioinformatics

Laura Sun (PID: A16923552)

AlphaFold Data Base (AFBD)

The EBI maintains the largest databaes of alphafold prediction models at: <https://alphafold.ebi.ac.uk>

From last class we saw that the PDB had 244,290 (October 2025)

The total number of protein sequences in UniProtKB is 199,579,901

Key point: this is a tiny fraction o f sequence space that has structural coverage (0.12%)

```
244290/199579901 * 100
```

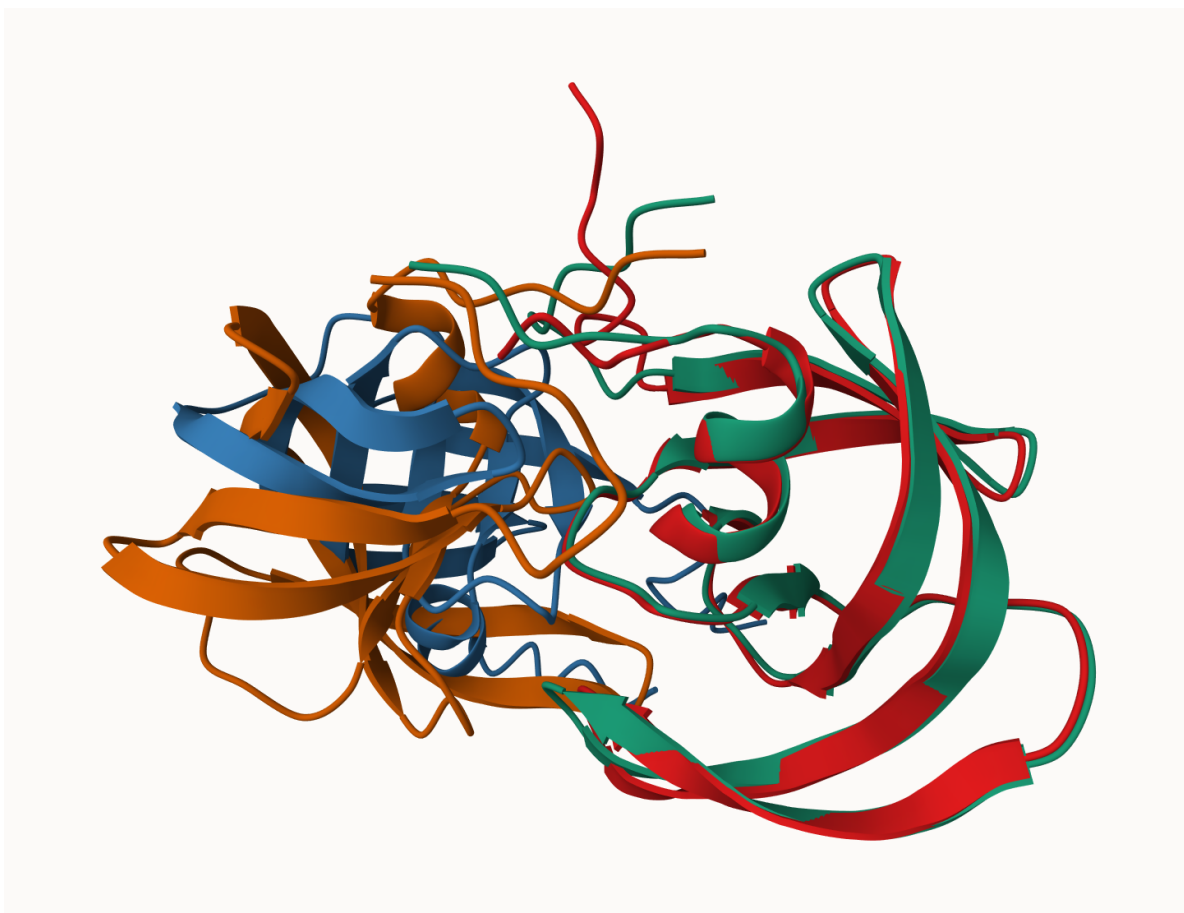
```
[1] 0.1224021
```

AFDB is attempting to address this gap...

There are two “quality scores” from AlphaFold. One for residues (i.e. each amino acid) called **pLDDT** score, the other called **PAE** Predicted Aligned Error measures the confidence in the relative position of two residues (i.e. a score for every pair of residues).

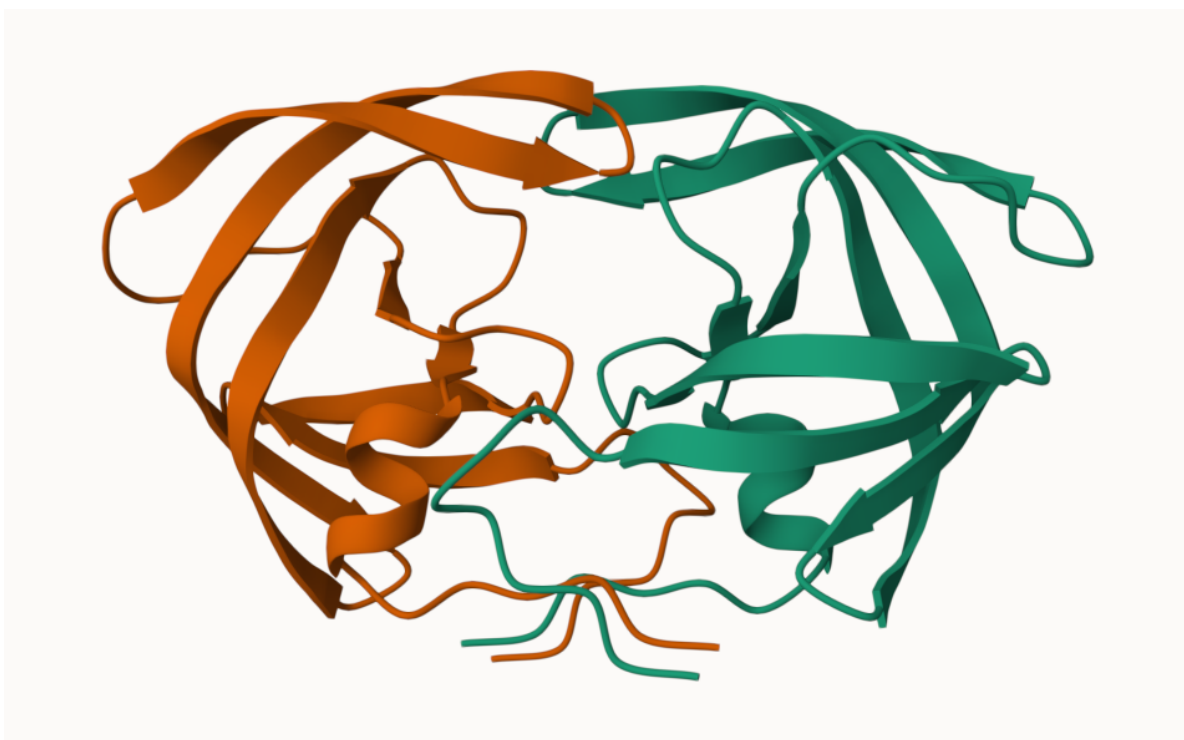
Generating your own structure predicitions

<https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/AlphaFold2.ipynb>

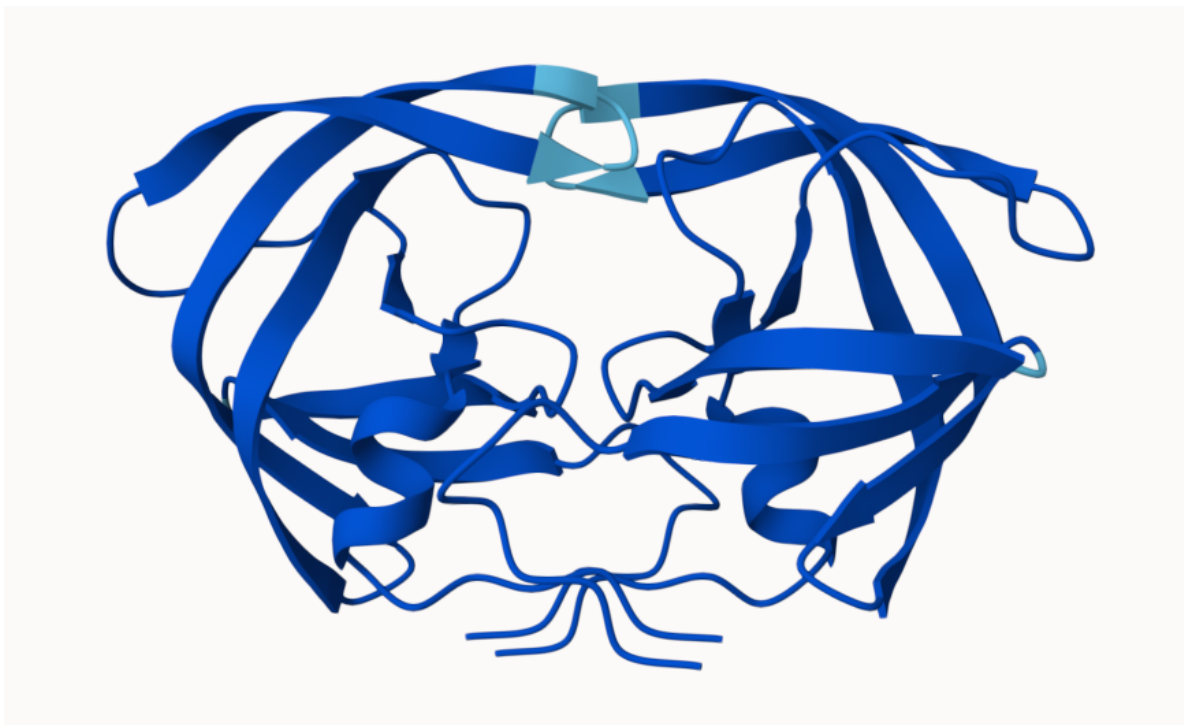


of all 5 superimposed

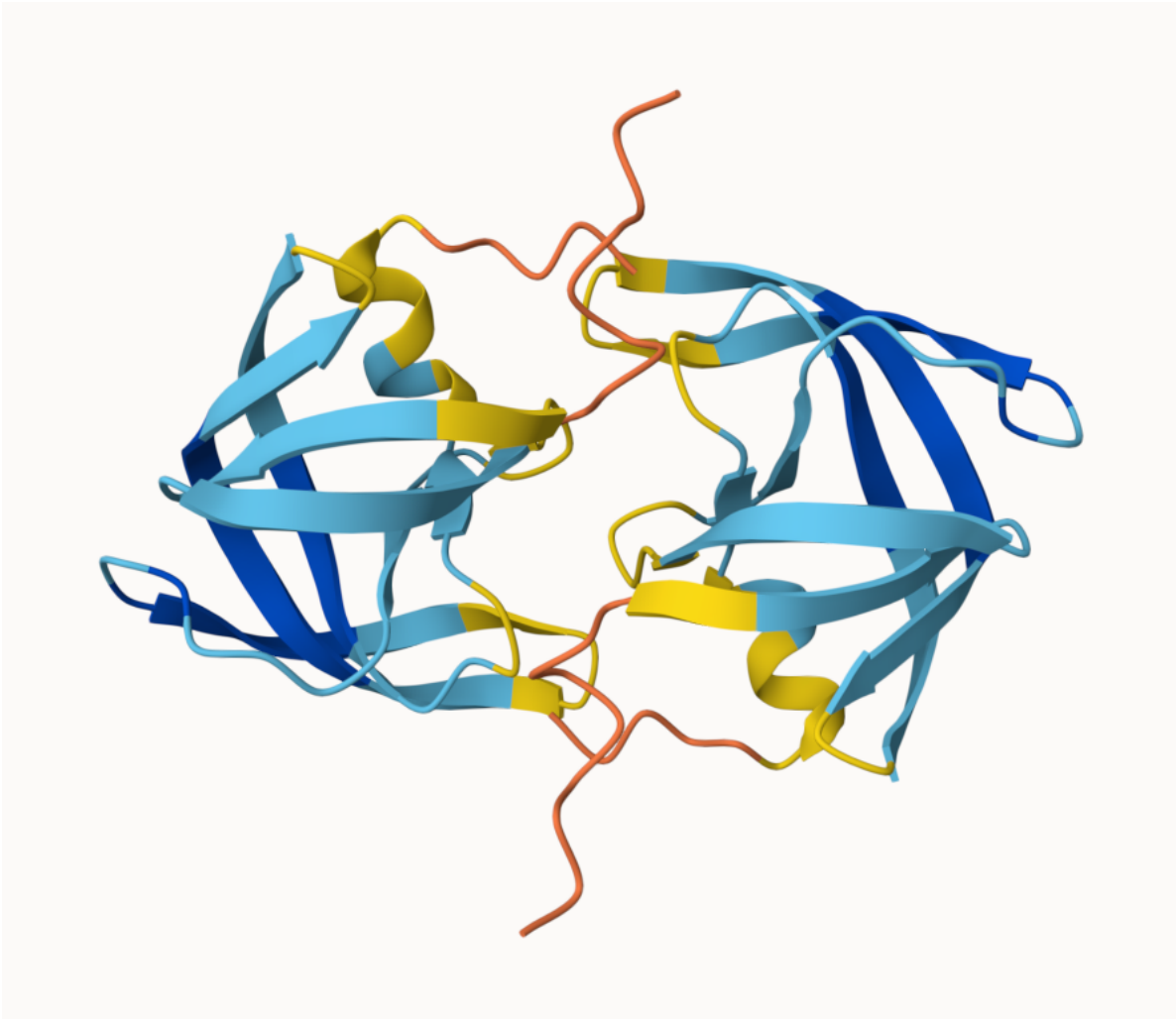
And the top ranked model colored by chain.



pLDDT score for model 1:



and model 5



Custom Analysis of Resulting Models in R

Read key result files into R. The first thing I need to know is what my results directory/folder is called (i.e. its name is different for every AlphaFold run/job)

```
results_dir <- "HIVPR_dimer_23119/"

pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)
```

```
# Print our PDB file names
basename(pdb_files)
```

```
[1] "HIVPR_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_2_seed_000.pdb"
[2] "HIVPR_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_4_seed_000.pdb"
[3] "HIVPR_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_1_seed_000.pdb"
[4] "HIVPR_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_5_seed_000.pdb"
[5] "HIVPR_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
library(bio3d)
```

```
m1 <- read.pdb(pdb_files[1])
m1
```

```
Call: read.pdb(file = pdb_files[1])
```

```
Total Models#: 1
```

```
Total Atoms#: 1514, XYZs#: 4542 Chains#: 2 (values: A B)
```

```
Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 0 (residues: 0)
```

```
Non-protein/nucleic resid values: [ none ]
```

```
Protein sequence:
```

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, calpha, call
```

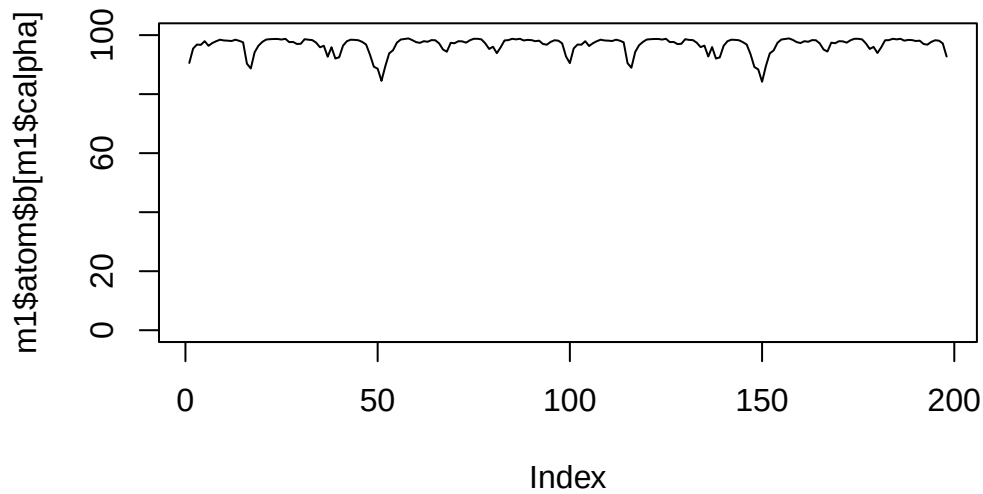
```
head(m1$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	-17.688	3.373	4.414	1	90.56
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	-17.859	2.959	3.020	1	90.56
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	-17.000	1.748	2.658	1	90.56

4	ATOM	4	CB	<NA>	PRO	A	1	<NA>	-17.422	4.195	2.230	1	90.56
5	ATOM	5	O	<NA>	PRO	A	1	<NA>	-16.078	1.391	3.404	1	90.56
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	-16.641	5.012	3.207	1	90.56

	segid	elemsy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	C	<NA>
5	<NA>	O	<NA>
6	<NA>	C	<NA>

```
plot(m1$atom$b[m1$calpha], typ="l", ylim=c(0,100))
```



Residue conservation from alignment file

Find the large AlphaFold alignment file

```
aln_file <- list.files(path=results_dir,
                      pattern=".a3m$",
                      full.names = TRUE)
aln_file
```

```
[1] "HIVPR_dimer_23119//HIVPR_dimer_23119.a3m"
```

Read this into R

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
```

```
[2] " ** Duplicated sequence id's: 101 **"
```

How many sequences are in this alignment

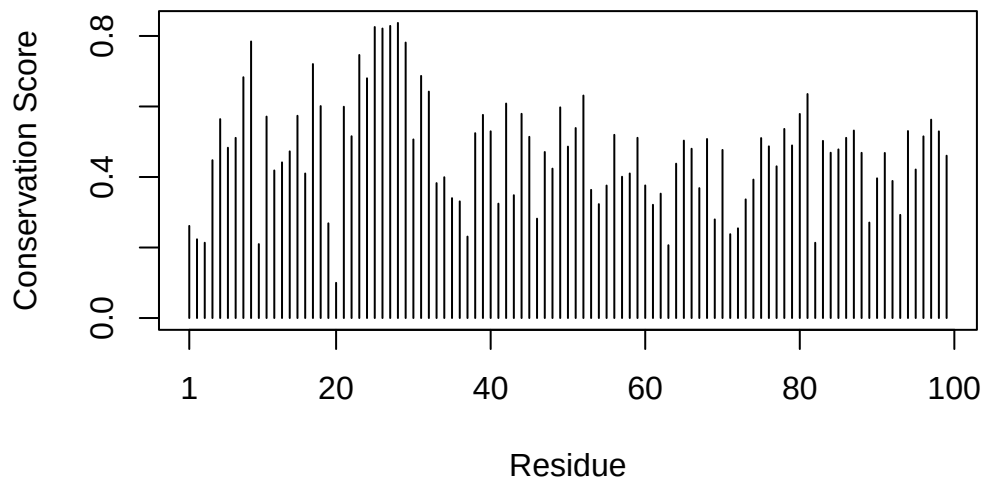
```
dim(aln$ali)
```

```
[1] 5397 132
```

We can score residue conservation in the alignment with the `conserv()` function.

```
sim <- conserv(aln)
```

```
plotb3(sim[1:99], ylab="Conservation Score")
```



```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```